

Delayed Verification in Multi-Agent LLM Systems: Stability of a Consensus Model and Coherence-Based Corrector Placement

Igor Itkin

Independent Researcher, Tel Aviv, Israel

ig.itkin@gmail.com

ORCID: 0009-0004-9513-8463

June 2026; revised September 16, 2026

Abstract

Multi-agent large language model (LLM) systems often rely on verifier and critic agents to suppress hallucinations, but verification is delayed. During this delay, false claims can propagate through the agent network. We model this process as delayed consensus on a graph with grounded corrector nodes. Spectral decomposition by the grounded Laplacian yields an exact stability threshold for the verification dose: correction that is too strong or too delayed can destabilize this linear model through an oscillatory mode. This does not establish a bounded nonlinear cycle. When communication and verification have the same delay, their gains add; at delay two the threshold for the dimensionless combined dose $\eta(\mu + \kappa)$ is the inverse golden ratio. A related coherence surrogate gives a supermodular placement objective: greedy assignment achieves at least $(1 - 1/e)$ of the optimal coherence reduction for a fixed corrector budget. This guarantee concerns the surrogate, not the mean-square error under constant random forcing. Experiments with an externally computed correction law show larger signed-error variability at longer effective lag in five of six tested models in an exploratory all-run analysis. A separate expanded factual study uses 400 new questions, corrected delay indexing and complete response logs. After semantic review, conservative completion bounds still permit both signs of the change in binary error amplitude, largely because many states contain abstentions. A post-hoc analysis finds lower abstention frequency at the longer lag on TruthfulQA, without establishing a change in error variability. Neither study establishes general factual stability; non-negativity alone does not rule out oscillations under persistent forcing.

Keywords: multi-agent LLM systems; hallucination cascade; verification latency; delay stability; grounded Laplacian; leader selection; corrector placement.

1 Introduction

Large language models hallucinate, and in multi-agent systems hallucination stops being a static property of one output and becomes a dynamic process: claims are exchanged, revised, and reused as context, so an unsupported claim from one agent can be amplified by others [3, 5, 4]. The phenomenon is sharp even inside a single model: Zhang et al. show that once an LLM commits to a wrong answer it generates further false claims to justify it (“hallucination snowballing”) even

when it can separately recognize them as wrong [8]. Approaches to improving responses include self-feedback [9], reinforced self-checking [10], and multi-agent debate [11, 12]. These mechanisms differ in their access to evidence: for example, Self-Refine uses the same LLM to generate, critique and refine an answer, without an oracle truth signal. Evidence-grounded verification is a distinct design choice.

Verification, however, has latency. A verifier reads a claim, retrieves evidence, and returns a correction only after several interaction steps; meanwhile the unverified claim has already propagated. Delayed negative feedback is the classic ingredient of oscillation and instability in control systems. We study the following question in a specified consensus model: *can the very act of verification, if delayed, destabilize the factuality it is meant to protect, and how should correctors be dosed and placed to avoid this?*

The agent-safety literature is moving from post-hoc analysis toward online monitoring of cascades (causal cross-channel monitoring [21], online failure auditing [20], temporal-graph anomaly detection [22]). These monitoring objectives are distinct from the stability of a controller that acts on delayed feedback. Quickest-change-detection theory studies detection delay under false-alarm constraints [26, 27, 28, 31]; here we analyze correction gain and delay in a postulated network recurrence. The single-agent detection problem is treated separately in our companion work [37].

Delayed institutional regulation of adaptive multi-agent systems provides related motivation [36]; that work reports a supercritical Hopf bifurcation in a different nonlinear model. That nonlinear conclusion does not transfer to our discrete linear recurrence. Existing dynamical accounts of LLM interaction include consensus models [42, 56], hidden-anchor deliberation [43], self-correction as feedback control [44], and models of factual-error propagation [3, 4, 5]. Our specific object is a discrete network recurrence with an explicitly delayed restoring term. Its stability threshold is a property of that recurrence; relating it to ordinary factual verification requires independent identification. Empirical debate failures [57] motivate this question but do not establish that this mechanism explains those failures.

This paper makes five contributions.

- **A model and a reduction.** We cast the corrected multi-agent loop as a delayed consensus with grounded correctors that the grounded Laplacian decouples into independent scalar delay recurrences (Lemma 1; Sections 3–4).
- **An exact verification-dose limit.** On the branch $0 < a < 1$, increasing the dimensionless dose $\beta = \eta\kappa$ first destabilizes the linear loop through a complex pair. The threshold decreases with delay and approaches the inverse golden ratio at delay two as $a \uparrow 1$; it is not a theorem of bounded nonlinear oscillation (Theorem 1, Proposition 2, Lemma 2, Corollary 1; Section 4).
- **Placement for a coherence surrogate.** The resolvent trace is supermodular, so greedy placement achieves at least $1 - 1/e$ of the optimal reduction in this metric. The guarantee does not transfer directly to mean-square error under constant forcing (Theorem 2; Section 5).
- **Two coupled delays.** We derive the unit-circle locus of the two-delay characteristic polynomial, including its singular cases. Equal delays give an exact combined-gain ceiling, equal to the inverse golden ratio at delay two (Theorem 3, Corollary 3; Section 6).
- **An empirical regime comparison.** Numeric-estimation debates driven by an external controller exhibit greater signed-error variability at longer lag in an exploratory all-run

comparison across several models. The archived experiment requires an indexing correction and separate conditional/all-run analyses. A new 400-question factual study records all responses and uses corrected indexing, but abstentions and residual scoring ambiguity leave the direction of its binary-amplitude contrast unidentified. A separate analysis distinguishes the frequency of each response outcome from its temporal variability; non-negativity alone does not establish stability (Remark 1; Section 7).

Project repository: <https://github.com/YehudaItkin/delayed-verification-llm>. The project contains numerical checks, archived trajectories and reanalysis scripts. Historical archives lack raw responses for some experiments, preventing independent reconstruction of their factual scores. The expanded study instead preserves full request/response logs, a frozen manifest, versioned semantic decisions and checksummed analyses (Section 7.2.1). The data layout and offline reproduction commands are given in Appendix Appendix E.

The theoretical argument has three steps: define the recurrence (Section 3), derive its scalar stability threshold (Section 4), and distinguish the placement objective from stability (Section 5). Section 6 adds communication delay. Section 7 then asks what synthetic and LLM experiments can establish about these models; its observed variability measures are not tests of asymptotic stability.

2 Related work

We relate empirical reliability studies of multi-agent LLMs to classical delayed-feedback control. The stability and placement ingredients have precedents in control theory; the contribution is their specialization to a stated verification-loop model, with explicit limits on its empirical interpretation.

On the LLM side, Jamshidi et al. track claim-level inconsistency across agent chains and report that deeper chains can reduce an explicit hallucination score while eroding factual accuracy [3]; their related work models collective hallucination and interventions [4]. Xie et al. model error cascades and study provenance-aware mitigation [5]. Evaluator preferences and unsafe content are studied as distinct propagation phenomena [6, 7]. These are already dynamical accounts, rather than a purely static literature. Debate can improve factuality [11, 12]; Liang et al. propose it as a remedy for degeneration during *self-reflection*, while also discussing judge bias [13]. Other work separates the contribution of voting from interaction [14], studies sycophantic consensus [15], and varies communication topology [16].

Related dynamical models include DeGroot-inspired consensus [42], hidden-anchor deliberation [43], and single-agent error-transition models [44]. Delay-free averaging need not make every agent’s opinion monotone: a cooperative two-node averaging matrix with diagonal 0.1 and off-diagonal 0.9 has disagreement multiplier -0.8 , giving damped alternation. Delay robustness also depends on the update rule. Yao and Li [2] retain a current self-state and delay normalized neighbor terms; our two-delay model delays the entire Laplacian term, including its diagonal. Their convergence result therefore cannot establish delay robustness of our gossip term. Indeed, with no verification, $p = 20/27$ and communication delay $d = 2$, the scalar polynomial is

$$z^3 - z^2 + 20/27 = (z + 2/3)(z^2 - 5z/3 + 10/9).$$

Its conjugate roots have squared modulus $10/9 > 1$, whereas instantaneous gossip has multiplier $7/27$. Thus delayed gossip alone can destabilize our model. Sections 4 and 6 distinguish instantaneous gossip from coupled communication and verification delays.

Detection and control address different questions. Failure attribution benchmarks [17, 18, 19] and online monitoring methods [20, 21, 22] locate or anticipate failures; we analyze a specified correcting action. Sequential detection provides related tools, including CUSUM and minimax delay criteria [26, 27, 28, 29, 30, 31, 32, 33], but no detection-delay result is proved here. Potential LLM monitoring signals include semantic uncertainty and self-consistency [23, 24, 25]; confidence-based early exit [45] and social-learning interventions [46] have different objectives. Our stability boundary specializes the discrete delay equation studied by Kuruklis [35].

The placement results adapt established network-control objectives. Clark et al. study supermodular *transient convergence error* [41]; steady-state variance under noise-corrupted leaders is treated directly by Mackin and Patterson [38]. These metrics should not be conflated with constant-forcing squared error. Our resolvent-trace proof and greedy reduction guarantee specialize this line of work [48, 49, 51]. Sherlock [47] instead studies budgeted verification in agentic workflow execution; that setting differs from stability and coherence in our symmetric consensus recurrence. The delay axis has a control precedent as well: Pirani et al. characterize the maximum-allowable-delay margin of leader-follower consensus through the grounded-Laplacian spectrum and turn it into a leader-placement centrality [50], but in continuous time and with a single delay; we add the discrete verification dose, the two coupled delays, and the LLM loop. The grounding signals themselves (claim verification and faithfulness [52, 53]) are standard.

3 Model

We abstract a round of multi-agent deliberation as follows. Each agent carries a scalar *belief* about a claim, represented by a real coordinate rather than a calibrated probability, and advances in rounds: agents revise their belief toward those of their neighbours (debate), while designated *corrector* agents, which have access to verified evidence, push beliefs back toward the ground truth. Two features of deployed systems drive the dynamics and are the focus of this paper. First, correction is *delayed*: a verifier reads a claim, retrieves evidence, and replies only after a latency, so it acts on a stale belief. Second, persistent faults can pull beliefs away from truth; we represent that influence by a constant additive forcing. The problem we study is then stated as: given the interaction graph, the verification delay, and the faulty forcing, (a) when does the loop converge to its generally biased equilibrium, and when does it lose stability, and (b) where should a limited budget of correctors be placed to reduce a specified coherence surrogate? The rest of this section makes the model precise; Sections 4–6 answer (a) and Section 5 answers (b).

Let $\mathcal{G} = (V, E)$ be a finite undirected weighted graph on $V = \{1, \dots, n\}$. Its adjacency matrix has $A_{ij} = A_{ji} \geq 0$, $A_{ii} = 0$, and an edge exactly when $A_{ij} > 0$; $D = \text{diag}(A\mathbf{1})$ and $L = D - A \succeq 0$ are its degree matrix and Laplacian. Here $\mathbf{1}$ denotes the all-ones vector, I the identity of the relevant dimension, and $B \succeq 0$ ($B \succ 0$) means that the symmetric matrix B is positive semidefinite (positive definite). We use tr for matrix trace and $\|\cdot\|_2$ for the Euclidean vector norm or its induced matrix norm. Agent i holds a scalar belief $b_{i,t} \in \mathbb{R}$ about a claim with ground truth b^* ; the error is $e_{i,t} = b_{i,t} - b^*$. A *corrector* set $R \subseteq V$ ($|R| = m$) consists of agents with access to verified evidence, held at truth ($e_{i,t} = 0$, $i \in R$), an ideal hard-pinning abstraction related to pinning control [55]. The remaining *free* agents $V_f = V \setminus R$ ($n_f = n - m \geq 1$) carry error $e_t \in \mathbb{R}^{n_f}$; a faulty subset injects a constant bias collected as a forcing $g \in \mathbb{R}^{n_f}$. Symmetry means reciprocal weights, a modeling assumption rather than a property inferred from the exchange of peer text in our experiments. It enables the real orthogonal decoupling of Lemma 1. Directed interaction matrices need not

be symmetric and may be non-normal; the real-spectrum thresholds below do not cover them in general. The hard pins remain at truth instantaneously. Separately, every free node receives the uniform delayed restoring term specified below. Its gain is not derived from the number or incident edges of the hard pins. This restoring signal is oracle-valued in the mathematical model; the experimental LLM verifier need not be correct. Content-level phenomena such as truth dilution and claim provenance are outside the scalar abstraction.

Free agents run consensus (step $\eta > 0$) plus a uniform *delayed verification* of gain $\kappa > 0$ and integer latency $\delta \geq 1$:

$$e_{t+1} = (I - \eta L_g) e_t - \eta \kappa e_{t-\delta} + \eta g, \quad (1)$$

where $L_g := L[V_f, V_f]$ is the principal submatrix of L on the free nodes (the *grounded Laplacian*). Reading (1) term by term: $(I - \eta L_g) e_t$ is the consensus (debate) step, in which each free agent moves toward its neighbours at rate η ; $-\eta \kappa e_{t-\delta}$ is the separate uniform restoring force toward truth, with *strength* κ and *delay* δ (it acts on the δ -step-old error); and ηg is the constant pull of the faulty agents. The verification strength κ and delay δ are the two control knobs, and everything that follows analyses (1) as a function of them.

The initial history $e_{-\delta}, \dots, e_0$ is arbitrary and finite, and updates start at $t = 0$. We call the homogeneous loop *asymptotically stable* if every such history converges to zero; for this finite-dimensional linear recurrence this is equivalent to every characteristic root lying strictly inside $|z| < 1$. With constant g , stability instead means convergence to the equilibrium after subtracting its constant history. A nonzero equilibrium error is therefore compatible with stability. At update $t \rightarrow t + 1$, delay δ means feedback from state $t - \delta$; the empirical immediate-feedback condition uses the extension $\delta = 0$.

The additive fault abstraction is not identical to eliminating fixed erroneous peers from a consensus graph. If external peers have fixed error vector f and weights $B \geq 0$ into the free nodes, their contribution is $\eta B f - \eta \text{diag}(B \mathbf{1}) e_t$. Thus exact elimination replaces L_g by $L_g + \text{diag}(B \mathbf{1})$ as well as setting $g = B f$. Omitting this diagonal term changes the stability problem. Equation (1) instead specifies L_g and additive g directly; we do not infer either operator from the fixed textual peers in the factual experiments.

To make the grounded Laplacian positive definite, we impose the following anchoring condition.

Assumption 1 (grounding reachability). Every connected component of the subgraph induced by V_f contains a node adjacent to R .

This condition is equivalent to $L_g \succ 0$. To see this, extend a free-node vector v by zeros on R to obtain $\tilde{v}$. Then

$$v^\top L_g v = \sum_{\{i,j\} \in E} A_{ij} (\tilde{v}_i - \tilde{v}_j)^2.$$

Under the condition, zero energy forces $v = 0$; without it, a nonzero constant on an unanchored free component has zero energy. We assume the condition for the hard-grounded stability results; Section 5 states separately when an ungrounded baseline is allowed. It is not necessary for stability of every corrected loop: the uniform verification term in (1) can also stabilize a zero Laplacian mode. For example, at $\delta = 1$ and $\eta \kappa = 1/2$, that mode has characteristic polynomial $z^2 - z + 1/2$, with roots $(1 \pm i)/2$ of modulus $1/\sqrt{2} < 1$. Moreover, for $\kappa > 0$, the operator $L_g + \kappa I$ is invertible even if L_g is singular. The unique equilibrium is $e_\infty = (L_g + \kappa I)^{-1} g$, which is truth when $g = 0$; convergence to it requires the stability conditions below.

4 Stability and the verification dose

Lemma 1 (grounded-Laplacian decoupling). *Let $L_g = Q \text{diag}(\mu_1, \dots, \mu_{n_f})Q^\top$, where Q is orthogonal and $0 < \mu_1 \leq \dots \leq \mu_{n_f}$ are the eigenvalues. Put $x_t = Q^\top e_t$, $\hat{g} = Q^\top g$. Since κI is scalar (hence commutes with L_g), (1) decouples into independent scalar recurrences*

$$x_{i,t+1} = a_i x_{i,t} - \eta\kappa x_{i,t-\delta} + \eta\hat{g}_i, \quad a_i := 1 - \eta\mu_i. \quad (2)$$

Proof. Multiply (1) by $Q^\top$ and use $Q^\top Q = I$ and $Q^\top L_g Q = \text{diag}(\mu_i)$. Both the instantaneous and delayed terms are then diagonal in the same coordinates, giving (2). $\square$

The decoupling is the crux of the analysis. An n_f -dimensional *delayed* network, ordinarily an awkward object, collapses into independent scalar modes, so the entire stability question reduces, after subtracting $x_{i,\infty} = \hat{g}_i/(\mu_i + \kappa)$, to the homogeneous scalar delay recurrence $x_{t+1} - a_i x_t + \beta x_{t-\delta} = 0$ (with $\beta = \eta\kappa$), one copy per eigenvalue μ_i of L_g . We analyse this scalar recurrence once and then quantify the result over the spectrum.

Proposition 1 (explicit $\delta = 1$ region). *For $\delta = 1$, mode (2) is asymptotically stable iff $\eta\kappa < 1$ and $\mu_i < 2/\eta + \kappa$.*

Proof. Characteristic $z^2 - a_i z + \eta\kappa$ with $a_i = 1 - \eta\mu_i$; Jury's conditions for $z^2 + c_1 z + c_0$ ($c_1 = -a_i$, $c_0 = \eta\kappa$) reduce to $\eta(\mu_i + \kappa) > 0$ (automatic), $\mu_i < 2/\eta + \kappa$, and $\eta\kappa < 1$. $\square$

Theorem 1 (networked stability). *Under Assumption 1, the homogeneous part of (1) is asymptotically stable iff $(a_i, \eta\kappa) \in \mathcal{D}_\delta$ for every $\mu_i \in \text{spec}(L_g)$, where spec denotes the eigenvalue set and*

$$\mathcal{D}_\delta = \{(a, \beta) \in \mathbb{R} \times (0, \infty) : \text{all roots of } z^{\delta+1} - az^\delta + \beta \text{ have } |z| < 1\}.$$

For $\delta = 1$ this is $\eta\kappa < 1$ and $\mu_{\max}(L_g) < 2/\eta + \kappa$. The positive-mode branch for larger delays is treated separately below.

Proof. By Lemma 1 the system is the direct sum of the scalar modes; it is stable iff each is. Apply Proposition 1. $\square$

For $\delta = 1$, stability requires both a verification gain below $1/\eta$ and a step size satisfying the fastest-mode constraint. Even single-step verification loses asymptotic stability at excessive gain. We next track the *onset* of instability for general delay δ on the non-negative mode branch used in Corollary 1.

Proposition 2 (oscillatory boundary for positive a). *Fix an integer $\delta \geq 1$ and the mode $x_{t+1} - ax_t + \beta x_{t-\delta} = 0$ with $\beta = \eta\kappa > 0$, $0 < a < 1$. As β grows from 0, the first roots to reach the unit circle are a simple complex-conjugate pair $e^{\pm i\theta_\star}$. The critical dose is*

$$a = \frac{\sin((\delta+1)\theta)}{\sin(\delta\theta)}, \quad \beta_c = \frac{\sin\theta}{\sin(\delta\theta)}, \quad \frac{\pi}{2\delta+1} < \theta < \frac{\pi}{\delta+1}, \quad (3)$$

where the branch contains a unique $\theta = \theta_\star$ for each $a \in (0, 1)$. The mode is asymptotically stable iff $0 < \beta < \beta_c$, so $\kappa_{\max}(a, \delta) = \beta_c/\eta$. The critical pair has angular frequency $\theta_\star$ radians per round (nominal period $2\pi/\theta_\star$ rounds). The sampled mode need not have an integer period.

Proof. See Appendix A. $\square$

At $a = 0$, the polynomial is $z^{\delta+1} + \beta$: all $\delta + 1$ roots have modulus $\beta^{1/(\delta+1)}$ and reach the unit circle simultaneously at $\beta = 1$. Thus the threshold extends to $a = 0$, but the assertion of a first isolated pair does not. Negative a requires a separate boundary analysis. For example, at $a = -1/2$, $\delta = 2$ and $\beta = 1/2$,

$$z^3 + \frac{1}{2}z^2 + \frac{1}{2} = (z+1)(2z^2 - z + 1)/2.$$

The first crossing is at $z = -1$; the other roots $(1 \pm i\sqrt{7})/4$ have modulus $1/\sqrt{2}$. For $0 < \beta < 1/2$, stability follows from $|z^2(z+1/2)| \geq 1/2 > \beta$ on $|z| = 1$ and Rouché's theorem. This case is outside Proposition 2.

Equation (3) specifies the boundary parametrically. Chebyshev polynomials give an equivalent algebraic equation and a direct proof of the monotonicity needed for a design rule.

Lemma 2 (Chebyshev form; the dose is monotone). *Let $U_0(c) = 1$, $U_1(c) = 2c$ and $U_{j+1}(c) = 2cU_j(c) - U_{j-1}(c)$ be the Chebyshev polynomials of the second kind. With $c = \cos \theta$ and $\sin(\delta\theta)/\sin \theta = U_{\delta-1}(c)$, the boundary (3) is algebraic,*

$$a = \frac{U_\delta(c)}{U_{\delta-1}(c)}, \quad \beta_c = \frac{1}{U_{\delta-1}(c)}, \quad c \in \left(\cos \frac{\pi}{\delta+1}, \cos \frac{\pi}{2\delta+1} \right) \quad (a \in (0, 1)),$$

with $\beta_c(a, 1) \equiv 1$, $\beta_c(0, \delta) \equiv 1$, and the exact value $\beta_c(a, 2) = \frac{1}{2}(\sqrt{a^2 + 4} - a)$ for $a \in [0, 1)$. For each $\delta \geq 2$ the dose is strictly decreasing in $a \in (0, 1)$; at $\delta = 1$ it is constant. For each fixed $a \in (0, 1)$, it is strictly decreasing in the integer delay $\delta \geq 1$, with $\beta_c < 1$ for $\delta \geq 2$. At $a = 0$ it remains 1 for every delay. The limit as $a \uparrow 1$ is $\beta_c(1, \delta) = 2 \sin(\pi/(4\delta + 2))$.

Proof. See Appendix B. □

On the branch $a = 1 - \eta\mu \in [0, 1)$, the ceiling does not increase as the delay δ grows and does not decrease as μ grows. Thus the slowest mode attains the binding constraint (all modes share the same ceiling when $\delta = 1$), which the corollary now pins down. Two remarks sharpen the claim. First, because (3) is a reparametrization of the exact, necessary-and-sufficient stability region of Kuruklis [35], $\kappa_{\max}(a, \delta)$ is not merely a sufficient bound but the *true* boundary on the stated branch; it is the discrete-time, delay-difference analogue of the continuous consensus delay margin $\tau^* = \pi/(2\lambda_{\max})$ of Olfati-Saber and Murray [54]. Second, the integer δ is the operative case: the proof in Appendix B establishes both monotonicity statements for arbitrary integer delays, with the boundary cases stated separately.

Corollary 1 (dose limit and binding mode). *Assume $\eta\mu_{\max}(L_g) \leq 1$ (so each $a_i \in [0, 1)$). Then the network dose limit is set by the slowest grounded mode, $\kappa < \kappa_{\max}(1 - \eta\mu_{\min}(L_g), \delta)$. For nested hard-corrector sets $R \subseteq R'$ with at least one free node remaining, Cauchy interlacing gives $\mu_{\min}(L_g(R')) \geq \mu_{\min}(L_g(R))$ and $\mu_{\max}(L_g(R')) \leq \mu_{\max}(L_g(R))$. At fixed η , adding hard correctors therefore preserves the assumed step-size condition and **does not lower the delay-induced dose ceiling**; the ceiling is unchanged at $\delta = 1$. In particular, reaching the dose ceiling loses asymptotic stability, and exceeding it gives a growing linear mode. Increasing delay can also cross the ceiling when the binding a is positive. The oscillatory loss of stability is the regime we call verification-induced oscillation.*

Proof. The largest $a_i = 1 - \eta\mu_i$ corresponds to $\mu_{\min}$, and Lemma 2 makes its threshold no larger than any other mode's. Proposition 2 and the $a = 0$ case give the network criterion. Interlacing of the nested principal submatrices gives the two eigenvalue inequalities and preserves $\eta\mu_{\max} \leq 1$. Applying the same threshold monotonicity proves the placement comparison. □

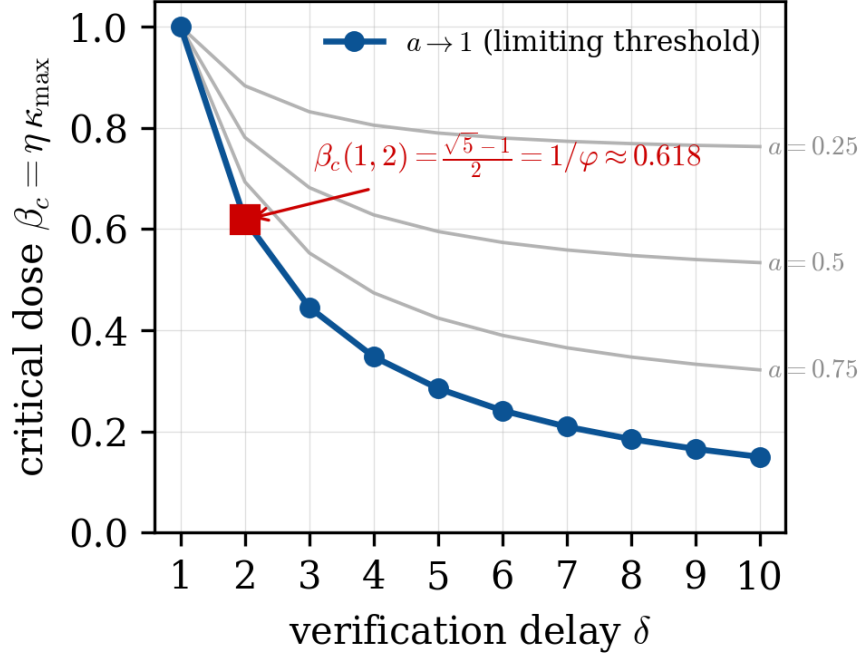


Figure 1: **The verification dose ceiling falls with delay.** Critical dose $\beta_c = \eta\kappa_{\max}$ versus verification delay δ for the limit $a \uparrow 1$ (blue) and three positive modes (grey); each scalar mode is stable below its threshold. The ceiling decreases with δ for these positive values of a ; at $\delta = 2$ the limiting blue curve equals the inverse golden ratio $(\sqrt{5} - 1)/2 \approx 0.618$ (red). More verification *latency* therefore forces a strictly smaller safe verification *strength*: the dose–delay tradeoff that the rest of the paper exploits.

5 Corrector placement for a coherence objective

Given a budget of k correctors, *where* should they go? This section first bounds the equilibrium error for the hard-grounded model, then defines a separate soft-pin placement problem on a fixed state space. We optimize a coherence surrogate, whose reduction is submodular. The greedy rule of Algorithm 1 achieves a fraction $1 - 1/e$ of the optimal reduction (Theorem 2). This is a guarantee for the chosen placement metric; it is not a guarantee for every measure of truth-tracking error.

The delay drops out of the equilibrium of (1):

$$(L_g + \kappa I) e_\infty = g \implies e_\infty(R) = (L_g + \kappa I)^{-1} g, \quad (4)$$

so *stability* is governed by $(\text{spec } L_g, \kappa, \delta)$ while *truth-tracking error* is governed by the resolvent $(L_g + \kappa I)^{-1}$ acting on the fault forcing. Placement R affects both quantities; these are distinct criteria rather than independently adjustable levers.

Corollary 2 (bounded steady-state error). *Whenever the loop is stable, $e_t \rightarrow e_\infty$ and the truth-tracking error obeys the resolvent bound*

$$\|e_\infty\|_2 \leq \frac{\|g\|_2}{\mu_{\min}(L_g) + \kappa}, \quad (5)$$

independent of the step size η and the delay δ . Thus the same $\mu_{\min}(L_g)$ that the dose limit (Corollary 1) makes the binding stability constraint also controls the residual error. Adding hard correctors never decreases $\mu_{\min}(L_g)$ (Cauchy interlacing), so the bound does not increase if the forcing on the remaining nodes is unchanged. Under the step-size condition of Corollary 1, the dose ceiling also does not decrease.

Proof. At equilibrium (1) gives $(L_g + \kappa I)e_\infty = g$; since $L_g \succ 0$ is symmetric, $\|(L_g + \kappa I)^{-1}\|_2 = 1/(\mu_{\min}(L_g) + \kappa)$, and submultiplicativity gives (5). Subtracting e_∞ gives the stable homogeneous recurrence, proving convergence for every initial history. Restricting an unchanged forcing to fewer free nodes cannot increase its norm. $\square$

The bound (5) shows that pinning helps, but not *which* nodes to pin: it sees only $\mu_{\min}(L_g)$. For the placement objective, fix a baseline hard-corrector set R_0 and $G = L[V \setminus R_0, V \setminus R_0]$. The set R_0 may be empty, as in the numerical example; $\kappa > 0$ still makes $G + \kappa I$ positive definite. In the rest of this section, R denotes additional *soft* pins on this fixed state space, each applying a finite restoring term $-we_i$ in the steady-state equation rather than imposing $e_i = 0$. We write u_i for the standard coordinate vector at node i , to distinguish it from a node's error. Their steady-state operator is

$$M(R) = G + W_R + \kappa I, \quad W_R = w \sum_{i \in R} u_i u_i^\top. \quad (6)$$

If g is sampled once, held constant in time, and has mean zero and covariance $\sigma^2 I$, then

$$\mathbb{E}\|M(R)^{-1}g\|_2^2 = \sigma^2 \operatorname{tr} M(R)^{-2}.$$

This differs from the *coherence* $H(R) = \operatorname{tr} M(R)^{-1}$ used in leader-selection objectives [48, 38]. For the separate, continuous-time, delay-free surrogate $dx = -M(R)x dt + dW_t$, where W_t is standard vector Brownian motion with independent unit-variance components, the stationary covariance P solves $MP + PM = I$. Hence $P = \frac{1}{2}M^{-1}$ and the stationary energy $\mathbb{E}\|x\|_2^2$ is $H(R)/2$. This noise model is not the constant forcing in (1), nor does it give the stochastic variance of the delayed discrete-time loop.

Both traces are non-increasing under added pins: each ordered eigenvalue of M is non-decreasing under the positive-semidefinite update, so both $\sum_j \lambda_j(M)^{-1}$ and $\sum_j \lambda_j(M)^{-2}$ are non-increasing. They need not rank two non-nested placements alike. For example, take the six-node graph with edges $\{02, 03, 04, 05, 12, 25, 34, 35\}$ (unit weights; nodes $0, \dots, 5$, with ij denoting $\{i, j\}$), $R_0 = \emptyset$, $\kappa = 1/2$, and $w = 2$:

Placement R	$H(R) = \operatorname{tr} M^{-1}$	$\operatorname{tr} M^{-2}$
$\{0, 1\}$	2.440088	1.640781
$\{1, 4\}$	2.383134	1.666253

The second placement improves coherence but worsens the static mean-square error. We therefore state the optimization problem explicitly for the coherence surrogate:

$$\min_{|R|=k} H(R) = \operatorname{tr} M(R)^{-1}, \quad (7)$$

where every node in the fixed set $V \setminus R_0$ is eligible and $1 \leq k \leq |V \setminus R_0|$. The optimization is over this static objective; it does not impose a delayed-loop stability constraint. Implementing the selected soft pins in a dynamic controller requires a separate stability check for that controller.

The coherence is supermodular, so its reduction $\rho(R) = H(\emptyset) - H(R)$ is monotone and submodular, and greedy achieves at least a fraction $1 - 1/e$ of the optimal *reduction* [51] (Theorem 2). The greedy step is cheap and interpretable, because its *marginal gain* has the Sherman–Morrison closed form

$$\Delta_i(R) := H(R) - H(R \cup \{i\}) = \frac{w \|M(R)^{-1} u_i\|^2}{1 + w u_i^\top M(R)^{-1} u_i} \geq 0, \quad (8)$$

a *resolvent centrality* that measures the coherence reduction from one additional pin and depends on the full inverse, rather than node degree alone. Greedy pins the node of largest Δ_i , updates $M(R)^{-1}$ by one rank-one step, and repeats (Algorithm 1). Here *submodular reduction* means diminishing returns: $\Delta_i(R) \geq \Delta_i(S)$ whenever $R \subseteq S$ and $i \notin S$. Equivalently, the decreasing cost H is supermodular.

Algorithm 1: Greedy corrector placement for problem (7): spend the budget one node at a time, each step pinning the node of largest marginal gain.

Input: fixed baseline Laplacian G , dose $\kappa > 0$, pin weight $w > 0$, budget k

Output: corrector set R with $|R| = k$

```

1  $R \leftarrow \emptyset$ 
2 for  $j \leftarrow 1$  to  $k$  do
3   foreach  $i \notin R$  do  $\Delta_i(R) \leftarrow w \|M(R)^{-1} u_i\|^2 / (1 + w u_i^\top M(R)^{-1} u_i)$  // (8)
4
5    $i^* \leftarrow \arg \max_{i \notin R} \Delta_i(R)$  // largest coherence reduction
6    $R \leftarrow R \cup \{i^*\}$ 
7 return  $R$ 
```

Theorem 2 (greedy coherence-reduction guarantee). *Let G be a fixed symmetric graph Laplacian or a principal submatrix of one, and let $\kappa, w > 0$. The coherence $H(R) = \text{tr } M(R)^{-1}$ is non-increasing and supermodular. Its reduction $\rho(R) = H(\emptyset) - H(R)$ is non-decreasing and submodular with $\rho(\emptyset) = 0$. For each integer budget $1 \leq k \leq |V \setminus R_0|$, the greedy rule satisfies*

$$\rho(R_{\text{greedy}}) \geq \left(1 - \left(1 - \frac{1}{k}\right)^k\right) \rho(R^*) \geq \left(1 - \frac{1}{e}\right) \rho(R^*), \quad (9)$$

where $R^* \in \arg \max_{|R|=k} \rho(R)$ is an optimal size- k set. In terms of the minimized objective, the last inequality is equivalent to

$$H(R_{\text{greedy}}) \leq \frac{H(\emptyset)}{e} + \left(1 - \frac{1}{e}\right) H(R^*).$$

This is not a multiplicative approximation factor for H , and gives no approximation factor for $\text{tr } M(R)^{-2}$.

Proof. See Appendix D. □

On a graph of three 5-cliques chained by unit-weight bridges, with $R_0 = \emptyset$, $\kappa = 0.3$ and $w = 12$ (Fig. 2), the rule is well separated from the two baselines at $k = 8$: greedy cuts the coherence $\text{tr } M(R)^{-1}$ from 9.1 to 2.3, against 2.9 for degree-based and 2.5 for random placement. Its first four picks are 4, 13, 8, 0 in the code’s zero-based indexing. The supermodular structure and the $(1 - 1/e)$ guarantee are *inherited* from leader selection in linear multi-agent systems [41, 48] and hold for the coherence objective (7); these comparisons do not measure the gap to an exact optimum or

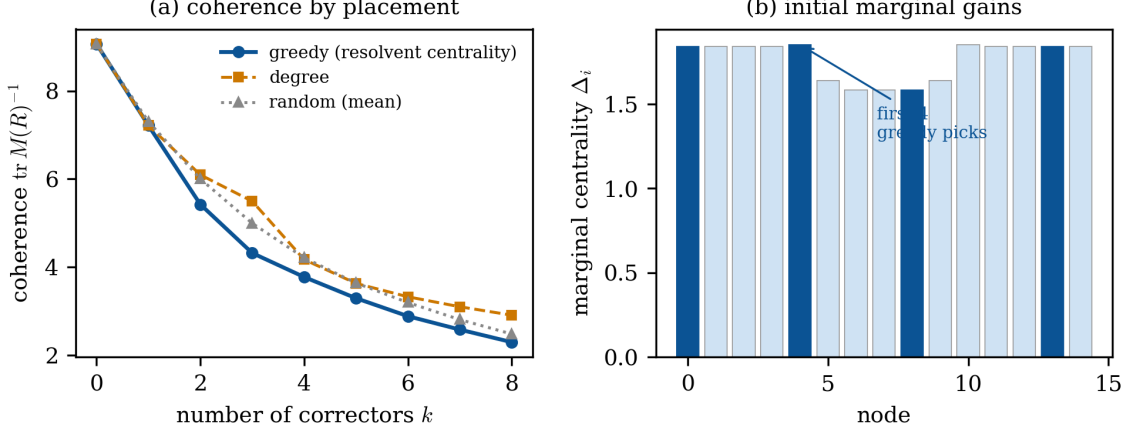


Figure 2: **Where to place correctors.** (a) On three 5-cliques chained by bridge edges, greedy selection by the resolvent centrality (8) is compared with degree-based and random placement (mean over 300 orders) using coherence $\text{tr } M(R)^{-1}$. An exact optimum is not plotted. (b) Initial marginal gains $\Delta_i(\emptyset)$ per node; the first four sequential greedy picks are marked in dark blue. Later picks use gains recomputed after the preceding pins.

establish a guarantee for a nonlinear error-propagation map. Hard grounding removes rows and columns: for nested corrector sets, the smallest grounded eigenvalue cannot decrease and the largest cannot increase. One way to see both directions is to extend a vector on the remaining nodes by zeros at the newly grounded nodes: its Rayleigh quotient is unchanged, while the minimization and maximization domains are restricted. For a three-node path, successively grounding its endpoints gives the spectra $\{0, 1, 3\}$, $\{(3 - \sqrt{5})/2, (3 + \sqrt{5})/2\}$, and $\{2\}$. This differs from a finite soft pin in (6), which adds a positive-semidefinite diagonal matrix on a fixed state space. These spectral facts do not establish an interior optimal budget k^* : such a claim needs an explicit cost or another competing constraint, beyond the fixed-budget placement problem studied here.

6 Two coupled delays

Let gossip carry integer latency $d \geq 0$ and verification integer latency $\delta \geq 1$:

$$e_{t+1} = e_t - \eta L_g e_{t-d} - \eta \kappa e_{t-\delta} + \eta g.$$

Lemma 1 still applies, giving per mode $x_{t+1} = x_t - \eta \mu x_{t-d} - \eta \kappa x_{t-\delta}$. A unit-circle root $\lambda = e^{i\theta}$ satisfies the pair

$$\cos \theta - 1 + \eta \mu \cos(d\theta) + \eta \kappa \cos(\delta\theta) = 0, \quad \sin \theta - \eta \mu \sin(d\theta) - \eta \kappa \sin(\delta\theta) = 0. \quad (10)$$

With $h = \max(d, \delta)$ and a supplied history $e_{-h}, \dots, e_0$, the characteristic polynomial is

$$P(z) = z^{h+1} - z^h + p z^{h-d} + q z^{h-\delta}, \quad p = \eta \mu, \quad q = \eta \kappa.$$

It generally has four terms when $d, \delta > 0$ are distinct. Related trinomial delay equations are studied in [34]; we derive the unit-circle locus for the polynomial above directly.

Theorem 3 (two-delay unit-circle locus). *For $d \neq \delta$ and $\theta \in (0, \pi)$ with $\sin((\delta - d)\theta) \neq 0$, a root $z = e^{i\theta}$ occurs exactly at*

$$p(\theta) = \frac{\sin(\delta\theta) - \sin((\delta + 1)\theta)}{\sin((\delta - d)\theta)}, \quad q(\theta) = \frac{\sin((d + 1)\theta) - \sin(d\theta)}{\sin((\delta - d)\theta)}.$$

At angles with $\sin((\delta - d)\theta) = 0$, the original two linear equations (10) must instead be checked for consistency; when consistent, they give additional lines in the (p, q) plane. The real-root loci are $p + q = 0$ for $z = 1$ and $p(-1)^d + q(-1)^\delta = 2$ for $z = -1$. These sets comprise the unit-circle locus, not a selection of its stability-bounding arcs. In each connected open cell of its complement, the number of roots inside the unit disk is constant; stable cells are those with all $h + 1$ roots inside. The origin itself is marginal, with a root at $z = 1$.

Proof. See Appendix C. □

For example, at $(d, \delta) = (1, 4)$ and $\theta = \pi/3$, the denominator vanishes and (10) reduces to $p - q = 1$. On that line,

$$P(z) = (z^2 - z + 1)(z^3 + qz + q).$$

At $q = 1/10$, the quadratic roots lie on the unit circle while the cubic roots lie strictly inside: on $|z| = 1$, $|qz + q| \leq 1/5 < |z^3|$, so Rouché's theorem applies. Thus singular cases cannot be discarded when constructing the boundary.

Corollary 3 (equal-delay combined-gain ceiling). *If $d = \delta = \tau \geq 1$, the two terms merge into a single lag at $a = 1$, $x_{t+1} - x_t + \eta(\mu + \kappa)x_{t-\tau} = 0$. For positive combined gain, it is stable iff*

$$0 < \eta(\mu + \kappa) < \beta_c(1, \tau) = 2 \sin \frac{\pi}{4\tau + 2}, \quad \text{e.g.} \quad \beta_c(1, 2) = \frac{\sqrt{5} - 1}{2} = \frac{1}{\varphi}.$$

For the network, this becomes $\eta(\mu_{\max}(L_g) + \kappa) < \beta_c(1, \tau)$: the largest grounded eigenvalue binds, unlike the instantaneous-gossip case of Corollary 1. Here $\varphi = (1 + \sqrt{5})/2$ is the golden ratio.

Proof. See Appendix C. □

This is an exact special case, not a universal ordering of delay configurations. Monotonicity in $a = 1 - \eta\mu$ for instantaneous gossip does not compare different positive gossip delays. In particular, at $\tau = 2$ the golden-ratio threshold applies to the *combined* gain $p + q$. For instantaneous gossip, the verification-only threshold $\beta_c(1 - p, 2)$ approaches $1/\varphi$ from above as $p \downarrow 0$.

A fixed-total-delay example shows why a comparison class is necessary. Fix $p = 1/10$ and $q = 98/125$. At $(d, \delta) = (1, 1)$ the polynomial is $z^2 - z + 221/250$; both roots have modulus $\sqrt{221/250} < 1$. With the same gains and total delay $d + \delta = 2$, the allocation $(0, 2)$ gives

$$z^3 - \frac{9}{10}z^2 + \frac{98}{125} = (z + \frac{7}{10})(z^2 - \frac{8}{5}z + \frac{28}{25}).$$

Its conjugate roots $(4 \pm 2i\sqrt{3})/5$ have modulus $\sqrt{28/25} > 1$. Thus equal delays are stable here while unequal delays are unstable. This example compares a fixed sum of delays; it does not establish an ordering for a fixed maximum delay or a fixed verification delay.

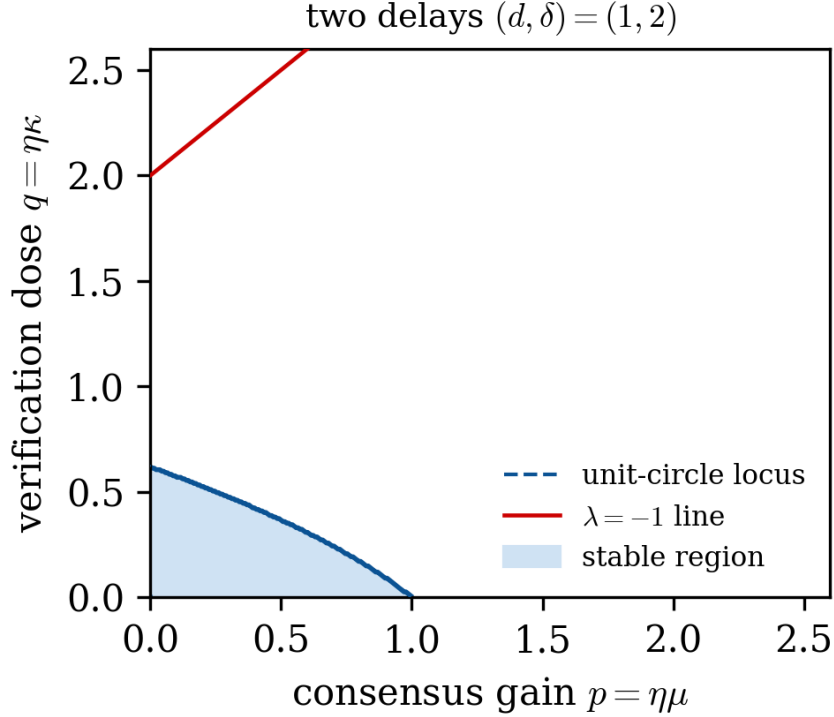


Figure 3: **Stability for a specified pair of delays.** Stability region (shaded) in the $(p, q) = (\eta\mu, \eta\kappa)$ plane for communication delay $d = 1$ and verification delay $\delta = 2$, computed by testing all characteristic roots. The regular unit-circle locus is dashed (Theorem 3); the $\lambda = -1$ line is red and here non-binding. The dose ceiling on the q -axis is $1/\varphi \approx 0.618$. This plot makes no general comparison across delay allocations.

7 Experiments and empirical scope

We ask three questions, with an archival and an expanded study for RQ2. RQ1: does the nonlinear loop lose stability near the predicted dose limit $\kappa_{\max}(\delta)$? RQ2: in a real grounded factual debate, how do final-step error and finite-window variability differ across verifier prompts and delays? RQ3: do LLM agents transmit an externally computed delayed correction signal, and how does their finite-window signed-error variability change with its lag?

The studies use different outcomes. RQ1 measures the late half peak-to-peak range of a simulated real-valued error. Historical RQ2 uses an automated entailment score; expanded RQ2 separates binary factual labels from abstentions and unresolved answers. RQ3 measures the temporal standard deviation of a normalized signed numeric error. We retain the original name “amplitude” where used in the records, but give its definition for each study; values are not directly comparable across these outcomes. Unless stated otherwise, temporal standard deviations are population standard deviations, dividing by the number of observed states.

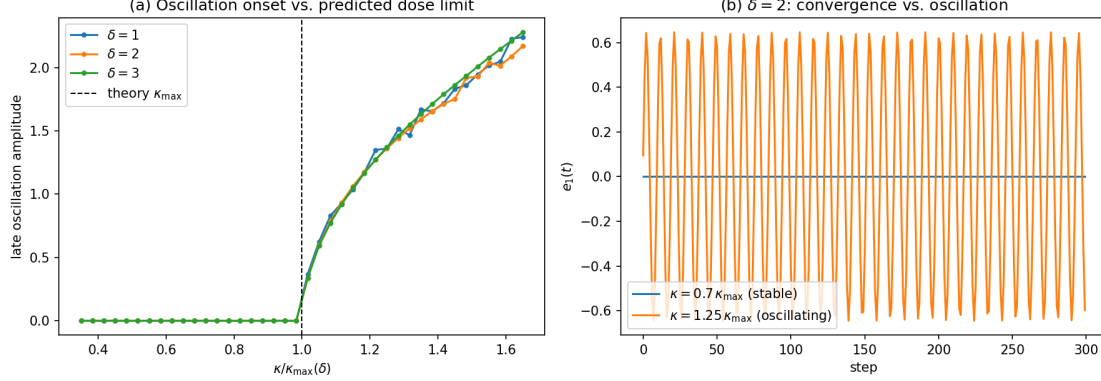


Figure 4: **Synthetic onset is consistent with the predicted ceiling at the sweep resolution.**

(a) oscillation amplitude collapses onto the predicted threshold $\kappa/\kappa_{\max} = 1$ for $\delta = 1, 2, 3$; (b) the $\delta=2$ trajectory converges below the ceiling and oscillates above it.

7.1 Onset at the predicted dose limit (RQ1)

We test whether the *linear* dose limit predicts onset in the *nonlinear* system with a saturating verifier,

$$e_{t+1} = (I - \eta L_g)e_t - \eta \kappa \tanh(e_{t-\delta}) + \eta g$$

($\tanh$ acts componentwise and $\tanh'(0) = 1$ matches the zero-state linearization). The ten-node graph has two hard pins (nodes 0, 5 in zero-based code indexing), leaving $n_f = 8$; it uses a random spanning tree with additional independent edges of probability 0.3 and seed 20260621. We set $\eta = 0.5/\mu_{\max}$ and a forcing 0.04 at the second free node, zero elsewhere, starting from an all-zero history. Each run has 4000 updates. Its late amplitude is half the largest nodewise peak-to-peak range over the final 400 states. The observed onset κ_{crit} tracks the predicted ceiling $\kappa_{\max} = \beta_c(1 - \eta\mu_{\min}, \delta)/\eta$:

δ	$\kappa_{\max}$ (theory)	κ_{crit} (nonlinear)	ratio
1	16.30	16.57	1.017
2	10.21	10.38	1.017
3	7.45	7.57	1.017

With nonzero forcing, zero need not be an equilibrium of the nonlinear loop; the comparison uses the zero-state linear reference and is not an exact local stability calculation at the forced equilibrium. The reported κ_{crit} is the first sampled dose whose late amplitude exceeds 10^{-3} . The sweep uses 40 equally spaced ratios $\kappa/\kappa_{\max}$ from 0.35 to 1.65: its spacing is 0.0333, and the samples bracketing the predicted boundary are 0.9833 and 1.0167. Thus the common ratio 1.017 is consistent with onset at the theoretical boundary at this resolution; it does not resolve a separate stabilizing shift due to saturation. At $\delta = 2$ and $\kappa = 1.25\kappa_{\max}$, the dominant nonzero Fourier bin of the first free node’s final 600 states gives a nominal period 9.38, compared with $2\pi/\theta_\star = 9.72$ at the linear boundary. This finite-window frequency estimate is not an exact period or a measurement at the threshold. An LLM front-end requires its own identification and cannot be assumed to preserve this update map.

7.2 Grounded factual debate: protocol-dependent outcomes (RQ2)

We reanalyze archived debates among instances of Qwen3.6-35B on selected PsiloQA [39] and TruthfulQA [40] questions. The archived filename suffix `tqa` refers to TruthfulQA: all 30 selected questions and gold answers match that dataset; the earlier TriviaQA label was erroneous. Selection used high cold-answer natural-language-inference (NLI) error within a candidate set, so these are difficulty-selected subsets rather than representative samples. Three free agents receive peer answers and a verifier note; grounded variants also supply four fixed wrong answers. Evidence is a Wikipedia passage for PsiloQA and a supplied gold-answer/alias statement for TruthfulQA. The verifier sees the first 600 characters of that evidence and is itself an LLM, not an oracle. The score $(1 - P(\text{entailment}) + P(\text{contradiction}))/2$ from `microsoft/deberta-large-mnli`¹ is a bounded error surrogate; no human calibration of this score is reported.

The 30-question PsiloQA sweep has 120 runs (18 states; weak/strong prompts and two stored delay labels). A separate 20-question sweep has 80 runs (36 states; strong/forceful prompts). In the latter, the final-step mean score is below 0.1 in 16/20 questions with the strong prompt and 8/20 with the forceful prompt, at either delay. This is an *endpoint success fraction*, not a convergence test. On the analogous TruthfulQA subset, the corresponding fractions are 4/20 for the strong prompt and 3/20 or 2/20 for the forceful prompt. In the separate ungrounded contrarian protocol, endpoint success is zero in all four cells. The later 80-run archive with saved majority strings has a mean normalized-string change rate of 90.8% after two warmup states. String changes need not be semantic changes, and the contrarian protocol also changes prompts and removes the faulty majority; it is not an isolated grounding ablation.

The F1 archive contains 180 runs per dataset: 15 questions, three delay labels, two forcing levels, two update temperatures, one repetition, and 20 states. The written plan instead specified five delays, three forcing levels, 30 questions, three repetitions, 24 states and a verifier-off control. Those missing conditions were not executed in the archived sweep. The analysis also substituted a two-delay Wilcoxon contrast for the planned ordered trend test. We therefore treat F1 as *exploratory*, not as execution of its confirmatory decision rule.

There is an additional indexing defect. For an update producing state t , the current state is $t - 1$; the archived F1 code uses the current state for label $r = 0$ and state $\max(0, t - r)$ otherwise. Labels $r = 0$ and $r = 1$ therefore both implement lag zero, and $r = 6$ implements lag five. The other factual harnesses similarly map labels 1, 4 to lags 0, 3, and labels 1, 6 to lags 0, 5. The corrected code records and uses theoretical indexing by default; these archived data remain unchanged. Furthermore, the nominal temperature-zero F1 condition used temperature 0.7 for initial answers, so it was not deterministic from initialization.

We recompute amplitude as the standard deviation of the final 10 mean-score states and pair contrasts by question. With four faulty peers, the label-6 minus label-0 mean differences and unadjusted one-sided Wilcoxon p -values are:

Dataset	Update temperature	Mean amplitude difference	p
PsiloQA	0	-0.0044	0.704
PsiloQA	0.7	+0.0023	0.433
TruthfulQA	0	+0.0317	0.124
TruthfulQA	0.7	+0.0350	0.094

¹Model card: <https://huggingface.co/microsoft/deberta-large-mnli>.

Each contrast has 15 question pairs. These tests do not reject at 0.05, even before correction for multiple comparisons; that is not evidence of equivalence or stability. Mean amplitude increases in three of the four contrasts. Without faulty peers, median amplitudes are often zero, but some trajectories still vary (maximum tail standard deviation 0.478). Variability at lag zero shows that delay is not necessary for variability in this protocol; it does not identify a unique cause. The archives omit per-agent answer text and verifier responses, preventing independent NLI rescoring. Non-negativity alone does not imply delay stability (Remark 1).

7.2.1 Expanded factual study with complete response logs

To address the small historical samples and missing response text, we ran a separate study on September 13–15, 2026 using the server model tag `qwen3.8:latest` through native Ollama. The frozen manifest records its artifact digest; the server tag is an operational identifier, not independent verification of a model-family or parameter-count claim. This study contains 200 new PsiloQA questions and 200 new TruthfulQA questions, excluding previously used questions. Selection used a deterministic hash ordering and source review before the new trajectories, without screening on new model responses or an NLI difficulty threshold. PsiloQA selection was from an exported pool of 4000 candidates, with at most one question per source article; TruthfulQA selection used the available unseen pool and an 80-character reference limit. These restrictions do not yield representative samples of all factual questions. The locally frozen protocol followed earlier pilots; it was not a public preregistration, and the earlier NLI-based power estimate does not transfer to this changed outcome measure.

Three free agents interact with four fixed erroneous peers. For each question, three initial answers are generated once and shared between the two conditions, with actual verification delays $\delta = 0, 5$. Each trajectory has 48 states, indexed $0, \dots, 47$; an update producing state t supplies the verifier with state $\max(0, t - 1 - \delta)$. Here $\delta = 0$ denotes immediate verification; communication has no added lag. Agent updates are synchronous, and the temperature is 0.7 for initialization and updates, with a temperature-zero verifier. Each request has a recorded seed; condition order is fixed by a hash. The verifier is another invocation of the same model, using up to 600 characters of frozen evidence. TruthfulQA evidence includes the reference answer: this is delayed reference feedback, not independent evidence retrieval. The primary grid holds the verifier on and the number of erroneous peers fixed; verifier-off and additional forcing conditions were not part of this expanded grid.

The output contract permits a short answer of at most 80 characters or an explicit abstention, with context 4096, generation budget 256 tokens and thinking disabled. This differs from the historical free-text verifier note, so the new study is not an isolated intervention on indexing. All 800 trajectories and 226800 requests completed; request-level parsing yielded 200065 valid answers, 26733 abstentions and two invalid answers, with no transport errors or length-limit terminations. Valid formatting does not ensure a complete or true statement. The request count includes verifier calls and differs from the number of stored agent states. Full logs reproduce prompts, seeds, synchronous states and delay indices; source-file hashes match the technical replay. Pilots and historical repetitions are excluded from the 400 questions.

Versioned scoring. The initial scorer uses a question-specific frozen catalogue, assigning M (reference match), C (reference contradiction), U (unresolved answer), A (abstention) or F (invalid format). Only M/C receive binary values 0/1. Of the unique pairs initially labelled U, all 3604 were

Dataset	Catalogue only	Semantic v1	Semantic v2
PsiloQA	$[-0.31336, 0.30800]$	$[-0.21964, 0.22103]$	$[-0.21110, 0.21280]$
TruthfulQA	$[-0.41625, 0.42362]$	$[-0.18397, 0.17875]$	$[-0.14170, 0.13446]$

Table 1: **Expanded factual study: conservative completion bounds for mean Δ .** All 200 questions per dataset are retained. Both signs remain possible under each scoring version. These intervals are not confidence intervals and do not establish absence of an effect.

read by the assisting language model in a packet hiding delay, trajectory position and frequency. The same assistant had already seen aggregate results, so this is neither independent human adjudication nor fully blind review. A source-conservative first pass, v1, assigned 1915 M, 662 C and 1027 U. A second pass, v2, assessed all remaining 1027 pairs by meaning, consulting external sources for selected disputed facts; it added 734 M and 113 C, leaving 180 U. Identical question–answer pairs retain identical labels in every occurrence.

Version v2 accepts recognizable spelling variants, some incomplete endings with unambiguous main content, and explicitly qualified descriptions of folklore. These post-hoc interpretive choices are not the official TruthfulQA evaluation, and the assistant’s judgments are not certified ground truth. Original M/C labels, abstentions and format failures were not relabelled. Both versions, their reasons and sources are preserved; decisions were frozen before their respective effect recalculations. Together the reviews classify 50149 of the 51670 initially unresolved stored agent states (97.1%), leaving 1521 U. This percentage measures scoring coverage, not accuracy.

Binary-amplitude contrast. For one question, let $y_{i,t,\delta} = 0$ for label M and 1 for label C; this binary outcome is distinct from the real-valued belief error $e_{i,t}$ in the model. It is undefined for A, U and F. For a fully scored or hypothetically completed trajectory let

$$A_\delta = \text{SD}_{t=24,\dots,47} \left(\frac{1}{3} \sum_{i=1}^3 y_{i,t,\delta} \right), \quad \Delta = A_5 - A_0,$$

using population standard deviation. We write $\text{SD}_{t \in T}(v_t) = [|T|^{-1} \sum_{t \in T} (v_t - \bar{v}_T)^2]^{1/2}$, where $\bar{v}_T = |T|^{-1} \sum_{t \in T} v_t$. For each question we compute the minimum and maximum A_δ over binary completions of unscored cells, then the outer interval $[A_{5,\min} - A_{0,\max}, A_{5,\max} - A_{0,\min}]$. Table 1 averages these endpoints over all 200 questions per dataset. For absent answers these completions are hypothetical: an abstention has no hidden factual truth value. Independently completing residual U occurrences can also be more permissive than enforcing one truth label across all occurrences of the same pair. Thus these are conservative outer bounds, not sharp semantic identification regions, confidence intervals or significance tests. They include neither sampling uncertainty nor possible adjudicator error. The finite optimization can be reproduced without enumerating all cell assignments: at each tail time, the possible erroneous-agent counts are consecutive integers from the known C count to that count plus the unscored count. A dynamic program starts at $(S, Q) = (0, 0)$ and, for each possible count c at the next time, adds $(S + c, Q + c^2)$. It therefore retains exactly the reachable sums of counts and squared counts. For $N = 24$ tail times and three agents, the variance is $Q/(9N) - S^2/(9N^2)$; taking the square roots of its minimum and maximum over reachable pairs gives the completion extrema. The implementation keeps only the smallest and largest Q for each S : future count choices do not depend on Q , and the final variance is increasing in Q at fixed S .

In the analysis tail, PsiloQA has 7781 A and 292 U cells, and TruthfulQA has 3510 A, 417 U

Dataset	Outcome frequency	$\delta = 0$ (%)	$\delta = 5$ (%)	Difference [95% interval], pp
PsiloQA	M	59.750	61.215	+1.465 [−0.597, 3.660]
	C	11.840	11.132	−0.708 [−2.389, 0.792]
	A	27.361	26.674	−0.687 [−2.076, 0.569]
	U	1.049	0.979	−0.069 [−0.271, 0.076]
	F	0.000	0.000	0.000 [0.000, 0.000]
TruthfulQA	M	78.090	79.062	+0.972 [−0.375, 2.162]
	C	7.750	7.812	+0.062 [−0.685, 0.845]
	A	12.799	11.576	−1.222 [−1.910, −0.550]
	U	1.347	1.549	+0.201 [−0.618, 1.110]
	F	0.014	0.000	−0.014 [−0.035, 0.000]

Table 2: **Expanded factual study: post-hoc outcome frequencies under v2.** Intervals concern paired changes in percentage points (pp), not separate condition means. All positions, including abstentions, remain in the denominator. A zero bootstrap interval for an unobserved event does not establish zero population probability.

and two F cells, each out of 28800 cells. Abstentions account for $11291/12002 = 94.1\%$ of cells without a binary score, not necessarily the same fraction of interval width. Fully scored tails in both conditions occur for only three PsiloQA and 15 TruthfulQA questions; restricting inference to those questions would select on model behavior. The direction of the original binary-amplitude contrast remains unidentified after the larger collection and review.

Separate analysis of observed outcomes. To describe behavior without assigning factual truth to an abstention, we subsequently defined $p_{k,t,\delta}$ as the fraction of the three agents receiving label $k \in \{M, C, A, U, F\}$. For each question we calculate its tail mean and population standard deviation, retaining every label and all positions in the denominator. In particular, $SD_t(p_{C,t,\delta})$ measures fluctuations in *labelled erroneous responses*; it is a different estimand from A_δ with undefined truth labels. Constant abstention can have frequency one and standard deviation zero.

For the mean paired difference of each outcome statistic, 20000 bootstrap resamples select whole, previously frozen question clusters with replacement, preserving both conditions and all questions within each selected cluster. Each resample draws the original number of clusters with equal probabilities; repeated clusters contribute repeated copies of their questions. The resampled statistic remains a mean over included questions, whose count can vary when cluster sizes differ. The reported interval is the empirical 2.5th to 97.5th percentile. There are 200 PsiloQA source-article clusters and 151 TruthfulQA question/family clusters, the latter with maximum size 11; resampling individual agent states would create pseudoreplication. We use the pairs cluster-bootstrap principle [1] and report pointwise percentile intervals. This is an exploratory analysis specified after reviewing earlier results, without multiplicity adjustment or a confirmatory hypothesis test. The intervals condition on the frozen labels and observed generation trajectories, assume independence between clusters, and do not account for adjudicator error or variation over new model-generation seeds.

TruthfulQA abstention frequency is lower at lag five by 1.22 percentage points, with a pointwise 95% interval [−1.91, −0.55]; this result is identical under v1 and v2 (Table 2). Fewer abstentions do not by themselves establish more correct answers. Indeed, the TruthfulQA M-frequency point difference changes from −0.326 percentage points under v1 to +0.972 under v2, and both intervals include zero. For the standard deviation of the labelled-error fraction, v2 gives changes +0.00282

(95% interval $[-0.00771, 0.01341]$) on PsiloQA and $+0.00211$ ($[-0.00485, 0.00915]$) on TruthfulQA; v1 likewise gives intervals spanning zero. These observations do not establish equivalence, stability or the proposed feedback mechanism. Full five-outcome means, standard deviations, paired results and both scoring versions are retained in the repository analysis files (Appendix Appendix E).

7.3 Externally controlled signed-error variability across models (RQ3)

We study a designed controller with LLM agents in its feedback loop, using eight author-curated numeric questions with fixed reference values and positive normalization scales (in the same units as each answer), recorded in `paper/remote/debate_v3.py`. The harness computes a suggestion $V_{i,t} = b_{i,t} - \alpha(b_{i,t-\ell} - b^*)$ outside the LLM, rounds it to one decimal place, and supplies it alongside peer estimates. The prompt instructs agents to follow the suggestion and peers and to use no outside knowledge. Thus the update law is imposed by the experimental protocol; it is not independently identified from ordinary factual verification.

An indexing audit changes the mapping between the stored delay labels and the model. In the archived harness, when producing state t , the current state is $t - 1$ but the stale state is $\max(0, t - r)$, where r is the saved label. Hence exact following would implement

$$e_{t+1} = e_t - \alpha e_{t-\ell}, \quad \ell = r - 1,$$

with the initial state repeated before time zero. The saved labels $r = 1, 6$ therefore correspond to lags $\ell = 0, 5$. For $\ell = 0$, stability requires $0 < \alpha < 2$; for $\ell = 5$, the ceiling is $2 \sin(\pi/22) \approx 0.285$. The contrast at $\alpha = 0.5$ still compares a stable and an unstable ideal controller. But $\alpha = 1.5, r = 1$ is also stable in the ideal controller, with damped alternating error; the claim that only the small-gain/small-delay cell is stable was incorrect. The corrected harness uses the theoretical indexing by default and records the convention. No new LLM runs under that corrected convention are reported here.

The numeric parser used in these archives extracts the last number-like substring; scientific notation and leading-dot decimals can be misread. The updated harness uses a strict finite-number parser. Raw responses were not archived, so the frequency of historical parsing errors cannot be established. Scores and trajectories were stored to three decimal places: recomputed amplitudes agree within the resulting 0.001 tolerance, while two movement flags lie at a rounding-ambiguous threshold. The analyses below retain the originally stored flags and amplitudes.

The stored experiment has 22 states per run and three repeated runs per question and cell. Its “amplitude” is the standard deviation of $e_t = (\bar{b}_t - b^*)/\text{scale}$, where $\bar{b}_t$ is the mean of the three free-agent estimates, over states $3, \dots, 21$; it measures finite-window variability, including transients. The overshoot statistic below requires both $\min_t e_t < -0.02$ and $\max_t e_t > 0.02$ over the stored trajectory. The separately stored `zc` field instead counts crossings of the trajectory’s own centred mean and is not used as a truth-overshoot statistic. Neither measure establishes persistent oscillation, a nonlinear limit cycle, or a bifurcation.

Using all runs and averaging within question, the Qwen3.6-35B amplitude is 0.0549 at $r = 1$ and 0.2648 at $r = 6$, a ratio of about 4.82. All eight question-level differences are positive, giving one-sided Wilcoxon $p = 0.00390625$. The same all-run direction and p occur for four other models; these five survive a Bonferroni adjustment across the six tested models ($6p = 0.0234375$). Mistral-Nemo gives $p = 0.0742$. These are exploratory all-run comparisons, not the movement-conditioned analysis specified in the original script.

Table 3: **All-run overshoot under the imposed controller.** Stored delay label r corresponds to effective lag $\ell = r - 1$. Percentages use all 24 trajectories per model and cell at $\alpha = 0.5$, with the two-sided truth-crossing threshold defined in the text.

Model	Developer	$r = 6$ ($\ell = 5$)	$r = 1$ ($\ell = 0$)
Qwen3.6-35B	Qwen	96%	0%
Qwen3-14B	Qwen	100%	0%
Mistral-7B	Mistral	100%	0%
Phi-4	Microsoft	96%	4%
Gemma-4-12B	Google	100%	0%
Mistral-Nemo-12B	Mistral	58%	0%

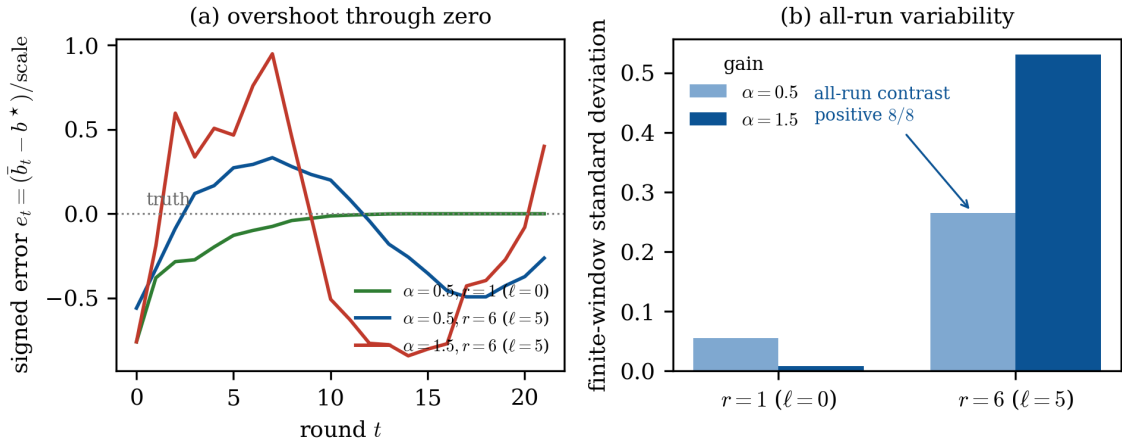


Figure 5: **Qwen3.6-35B under an externally computed controller.** (a) Selected signed-error trajectories from the saved runs. (b) All-run mean finite-window standard deviation for each gain and stored label r ; the actual lag is $\ell = r - 1$. At $\alpha = 0.5$, all eight question-level means increase from $r = 1$ to $r = 6$. This is the exploratory all-run comparison; the movement-conditioned result is reported separately. Neither panel classifies asymptotic stability from finite observations.

The specified movement filter retains runs with range greater than 0.15 over states $3, \dots, 21$. Applying it separately within each cell leaves 6 paired questions for Qwen3.6 ($p = 0.015625$), 7 for Gemma ($p = 0.0078125$), 5 for Phi-4 ($p = 0.03125$), and 7 for Nemo ($p = 0.0546875$). Qwen3-14B and Mistral-7B have no retained short-lag runs, so that paired comparison is undefined for them. The five-model all-run conclusion therefore cannot be presented as five successful tests of the movement-conditioned plan. The filter itself selects on an outcome of the intervention, so the conditional and unconditional comparisons answer different questions.

These results demonstrate that agents can transmit a delayed correction signal supplied by external code, with model-dependent deviations from exact following. They do not establish that natural verification has this update law, that deployed agents are necessarily more stable, or that the finite observed trajectories approach a bounded limit cycle. The stored metadata also says three faulty peers, whereas the available driver uses the function default of one; the executed count cannot be resolved from the aggregate trajectories alone. The fields named “seed” are repetition indices in this driver, which does not pass a random seed to the backend.

8 Discussion

In the postulated recurrence, verification is a delayed control action with a finite gain margin. Corollary 1 says a verifier can be *too* aggressive or *too* slow; Corollary 3 gives the exact combined-gain ceiling when communication and verification latencies coincide. These results give a model-specific reason to check feedback dynamics when changing verifier strength or latency. Placement affects both stability and tracking in the stated model. Our greedy guarantee concerns coherence reduction, while the stability comparison for nested hard-corrector sets follows from the slowest grounded mode. This complements online cascade *detectors* [21, 20, 22] with a *controller*-side stability guarantee that detection alone does not provide.

Remark 1 (non-negative error and absorption). The empirical contrast in Section 7 does not follow from the sign of the error measure alone. Consider the illustrative projected recurrence

$$p_{t+1} = \max(0, p_t - \alpha \sigma(p_{t-\delta}) + g),$$

with $\alpha > 0$, integer $\delta \geq 0$, non-negative initial history, and continuous σ satisfying $\sigma(0) = 0$ and $\sigma(p) > 0$ for $p > 0$. If $g = 0$, the sequence for $t \geq 0$ is non-increasing and converges to zero. Indeed, a positive limit ℓ would leave an asymptotically positive decrement $\alpha\sigma(\ell)$, contradicting convergence. Once zero is reached it is absorbing, including while the delayed history is still positive.

If $g > 0$, however, an all-zero history produces $p_{t+1} = g$, and non-negative periodic orbits are possible. For example, with $\alpha = 1$, $\delta = 1$, $g = 1$ and $\sigma(p) = \max(-2, \min(p, 2))$, the history $(p_{-1}, p_0) = (0, 0)$ generates $0, 0, 1, 2, 2, 1, 0, 0, \dots$, a period-six orbit. With $\sigma = \tanh$, the parameters $\alpha = 1.5$, $\delta = 1$, $g = 0.2$ also produce recurrent non-negative oscillations in the finite numerical check; exact periodicity is not proved for that example. Thus projection removes signed overshoot but does not establish absence of oscillation under forcing. The factual-QA observations should be interpreted as protocol-specific evidence; the projected recurrence is not an identified model of those experiments. Likewise, loss of linear stability in the signed model alone does not prove existence of a bounded nonlinear limit cycle.

9 Limitations

Several limitations bound the scope. The analysis is exact for the postulated linear recurrence. With constant forcing, its stable equilibrium is generally biased away from truth. Interpreting the recurrence as a local model of a nonlinear LLM loop requires an independently identified equilibrium and Jacobian; the synthetic onset comparison does not supply that identification. Unlike the dose limit, the placement rule is so far validated only on the coherence surrogate, not with LLM agents. Its approximation guarantee does not cover static mean-square error or the stochastic variance of the delayed loop. We model constant additive fault forcing, hard pins and a separate delayed restoring term; the placement surrogate uses finite soft pins. Byzantine faults and edge-level (message-filtering) correctors are future work. For general unequal delays, Theorem 3 gives candidate boundary loci, including singular cases; it does not provide a closed-form classification of every stable cell or a universal worst delay allocation. The eigen-decoupling assumes a *symmetric* (reciprocal) interaction graph. Directed grounded Laplacians can be non-normal, but directedness alone does not imply non-normality. For example, $2I - P$, where P is a three-cycle permutation matrix, is nonsymmetric and normal: it describes a directed cycle with an additional unit truth-pinning edge at each free

node. It has complex eigenvalues, so the real-mode formulas do not directly apply even in this normal case. Non-normal cases can additionally exhibit transient amplification. Directed extensions require separate stability and transient analyses; the symmetric thresholds are not guaranteed conservative.

The controlled numeric-estimation study is limited by eight questions, three repetitions per cell, a short observation window, the imposed correction law, the legacy indexing and metadata discrepancies, and the mismatch between movement-conditioned and all-run analyses. Its finite-window standard deviation includes transients; a stronger validation would identify dynamics independently and test persistence at longer horizons. Neither grounded factual-QA study quantitatively identifies the proposed mechanism. The historical AR diagnostic used the stored delay labels as actual lags and classified a high constant final error as observed instability. The previously quoted precision and recall were classification metrics, not coefficient-recovery accuracy. We withdraw that interpretation. At effective lag zero, current and delayed regressors are identical, so their separate coefficients are unidentifiable even with unlimited samples. The lag-aware diagnostic now reports only the combined coefficient there; short, quantized, often constant trajectories also limit inference at positive lag. The absorption argument applies only to the unforced projected model (Remark 1).

The expanded factual study supplies complete logs and corrected indexing, but tests one server model artifact, two delays, one faulty-peer count and a single trajectory per question and condition. Its format and evidence differ from the historical protocol, and its primary grid has no verifier-off control. Source review does not certify every reference, and post-hoc same-assistant semantic review is not independent human validation. The 80-character limit can leave unfinished statements even without a token-limit termination. Repeated questions from related TruthfulQA families motivate cluster resampling, but the chosen clustering does not guarantee independence across families. The expanded sample size is not a demonstrated power calculation for the revised metric. Stronger claims require a prospectively specified outcome that addresses abstentions, an explicit effect size of interest, score validation, replication on another model and longer horizons; the existing data should not be reclassified or selectively extended until a desired sign appears.

Finally, heterogeneous correction gains do not automatically preserve the modal reduction. For a diagonal gain matrix K , independent grounded-Laplacian modes require a common eigenbasis (equivalently, $[K, L_g] = 0$ for these symmetric matrices). This fails in general. Common stochastic delays with scalar gains may preserve the spatial modes, but require a time-varying stability analysis rather than a fixed characteristic polynomial. Coupling correction to an online change detector likewise remains outside the present results.

10 Conclusion

We modeled the multi-agent LLM verifier loop as a delayed consensus over a graph with grounded corrector nodes, and derived exact thresholds showing that verification carries a stability budget in this model. Excessive correction gain can cause loss of linear stability. When communication and verification delays coincide, the ceiling for the dimensionless combined dose $\eta(\mu + \kappa)$ is the inverse golden ratio at delay two. This does not make equal delays universally worst. For a fixed corrector budget, greedy placement guarantees at least $1 - 1/e$ of the optimal coherence reduction; this does not imply optimal static mean-square error.

The synthetic loop brackets onset near the linear prediction at the sweep resolution. The LLM experiment demonstrates transmission of an externally specified correction signal. Correcting the

history indexing maps its stored labels 1, 6 to effective lags 0, 5; the exploratory all-run comparison supports larger variability at the longer lag in five of six models. This does not establish the same law for natural factual verification or persistent nonlinear oscillation. The historical factual-QA contrasts are inconclusive and have small selected samples, aliased delay labels and departures from the planned controls. The separate expanded study addresses sample count, delay indexing and missing response logs, but its 400 questions still leave both signs of the binary-amplitude contrast possible under conservative completion bounds. Abstentions dominate the remaining unscored cells. A post-hoc outcome analysis finds lower abstention frequency at lag five on TruthfulQA, identically under both scoring versions; intervals for the change in labelled-error variability span zero. This does not establish greater accuracy, absence of an amplitude effect or general factual stability.

Within the assumed recurrence, dose is limited by stability, while a placement objective must specify the error or noise model it represents.

Acknowledgments

The author thanks Ilya Makarov for valuable feedback on the manuscript.

References

- [1] A. C. Cameron, D. L. Miller. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2) (2015). doi:10.3368/jhr.50.2.317.
- [2] L. Yao, A. Li. Convergence of time-delayed opinion dynamics with complex interaction types. *arXiv:2501.12219* (2025).
- [3] S. Jamshidi, A. Moradi Dakhel, K. W. Nafi, F. Khomh. Hallucination cascade: analyzing error propagation in multi-agent LLM systems. *arXiv:2606.07937* (2026).
- [4] S. Jamshidi. Collective hallucination in multi-agent LLMs: modeling and defense. *arXiv:2606.07941* (2026).
- [5] Y. Xie et al. From spark to fire: modeling and mitigating error cascades in LLM-based multi-agent collaboration. *arXiv:2603.04474* (2026).
- [6] Z. Liu. Contagion networks: evaluator preference propagation in multi-agent LLM systems. *arXiv:2606.20493* (2026).
- [7] B. Yan et al. PropGuard: safeguarding LLM-MAS via propagation-aware exploration and remediation. *arXiv:2605.16346* (2026).
- [8] M. Zhang, O. Press, W. Merrill, A. Liu, N. A. Smith. How language model hallucinations can snowball. *ICML 2024*. *arXiv:2305.13534*.
- [9] A. Madaan et al. Self-Refine: iterative refinement with self-feedback. *NeurIPS 2023*. *arXiv:2303.17651*.
- [10] Z. Li et al. MARCH: multi-agent reinforced self-check for LLM hallucination. *arXiv:2603.24579* (2026).

- [11] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, I. Mordatch. Improving factuality and reasoning in language models through multiagent debate. ICML 2024. arXiv:2305.14325.
- [12] J. C.-Y. Chen, S. Saha, M. Bansal. ReConcile: round-table conference improves reasoning via consensus among diverse LLMs. ACL 2024. arXiv:2309.13007.
- [13] T. Liang et al. Encouraging divergent thinking in LLMs through multi-agent debate. EMNLP 2024. arXiv:2305.19118.
- [14] H. K. Choi, X. Zhu, S. Li. Debate or vote: which yields better decisions in multi-agent LLMs? NeurIPS 2025. arXiv:2508.17536.
- [15] A. Wynn, H. Satija, G. Hadfield. Talk isn’t always cheap: understanding failure modes in multi-agent debate. ICML 2025 MAS Workshop. arXiv:2509.05396.
- [16] Y. Li et al. Improving multi-agent debate with sparse communication topology. arXiv:2406.11776 (2024).
- [17] X. Liu, X. Yang, Z. Li, P. Li, R. He. AgentHallu: benchmarking automated hallucination attribution of LLM-based agents. arXiv:2601.06818 (2026).
- [18] S. Zhang et al. Which agent causes task failures and when? On automated failure attribution of LLM multi-agent systems. ICML 2025. arXiv:2505.00212.
- [19] D. Deshpande et al. TRAIL: trace reasoning and agentic issue localization. arXiv:2505.08638 (2025).
- [20] B. Zhang et al. AgentForesight: online auditing for early failure prediction in multi-agent systems. arXiv:2605.08715 (2026).
- [21] K. Venkatesh, J. Isbarov, S. Amin, M. Kantarcioglu, J. Cui. CASPIAN: online detection and attribution of cascade attacks in LLM multi-agent systems via cross-channel causal monitoring. arXiv:2605.19240 (2026).
- [22] J. Zhou, L. Wang, X. Yang. GUARDIAN: safeguarding LLM multi-agent collaborations with temporal graph modeling. NeurIPS 2025. arXiv:2505.19234.
- [23] L. Kuhn, Y. Gal, S. Farquhar. Semantic uncertainty: linguistic invariances for uncertainty estimation in NLG. ICLR 2023. arXiv:2302.09664.
- [24] P. Manakul, A. Liusie, M. J. F. Gales. SelfCheckGPT: zero-resource black-box hallucination detection for generative LLMs. EMNLP 2023. arXiv:2303.08896.
- [25] L. Huang et al. A survey on hallucination in large language models. ACM TOIS (2025). doi:10.1145/3703155. arXiv:2311.05232.
- [26] E. S. Page. Continuous inspection schemes. *Biometrika* 41 (1954) 100–115.
- [27] G. Lorden. Procedures for reacting to a change in distribution. *Ann. Math. Statist.* 42 (1971) 1897–1908.
- [28] M. Pollak. Optimal detection of a change in distribution. *Ann. Statist.* 13 (1985) 206–227.

- [29] G. V. Moustakides. Optimal stopping times for detecting changes in distributions. *Ann. Statist.* 14 (1986) 1379–1387.
- [30] T. L. Lai. Information bounds and quick detection of parameter changes in stochastic systems. *IEEE Trans. Inf. Theory* 44 (1998) 2917–2929.
- [31] V. V. Veeravalli, T. Banerjee. Quickest change detection. In *Academic Press Library in Signal Processing* 3 (2014) 209–255. arXiv:1210.5552.
- [32] L. Xie, S. Zou, Y. Xie, V. V. Veeravalli. Sequential (quickest) change detection: classical results and new directions. *IEEE J. Sel. Areas Inf. Theory* 2 (2021) 494–514. arXiv:2104.04186.
- [33] A. Tartakovsky, I. Nikiforov, M. Basseville. *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Chapman & Hall/CRC (2014).
- [34] M. M. Kipnis, R. M. Nigmatullin. Stability of the trinomial linear difference equations with two delays. *Autom. Remote Control* 65(11):1710–1723 (2004).
- [35] S. A. Kuruklis. The asymptotic stability of $x_{n+1} - ax_n + bx_{n-k} = 0$. *J. Math. Anal. Appl.* 188 (1994) 719–731.
- [36] I. Itkin. Delayed repression and emergent instability in adaptive multi-agent systems. arXiv:2605.30392 (2026).
- [37] I. Itkin. Quickest detection of hallucination onset: delay bounds and learned CUSUM statistics. arXiv:2606.12476 (2026).
- [38] E. Mackin, S. Patterson. Submodular optimization for consensus networks with noise-corrupted leaders. arXiv:1712.08212 (2017).
- [39] E. Rykov et al. When models lie, we learn: multilingual span-level hallucination detection with PsiloQA. Findings of EMNLP (2025), 11663–11682. doi:10.18653/v1/2025.findings-emnlp.626.
- [40] S. Lin, J. Hilton, O. Evans. TruthfulQA: measuring how models mimic human falsehoods. arXiv:2109.07958 (2021).
- [41] A. Clark, B. Alomair, L. Bushnell, R. Poovendran. Minimizing convergence error in multi-agent systems via leader selection: a supermodular optimization approach. *IEEE Trans. Autom. Control* (2014). arXiv:1306.4949.
- [42] I. Yazici, M. Kayaalp, S. Taga, A. H. Sayed. Opinion consensus formation among networked large language models. ICASSP 2026. arXiv:2601.21540.
- [43] A. Pokharel, R. Dantu. Hidden anchors in multi-agent LLM deliberation. arXiv:2606.19494 (2026).
- [44] A. Liu, J. Meng. Self-correction as feedback control: error dynamics, stability thresholds, and prompt interventions in LLMs. arXiv:2604.22273 (2026).
- [45] T. Xu et al. Unveiling the entropy dynamics of chain-of-thought reasoning. ICML 2026. arXiv:2606.02020.

- [46] A. Jain, V. Krishnamurthy. Interacting large language model agents: interpretable models and social learning. arXiv:2411.01271 (2024).
- [47] Y. Ro, H. Qiu, Í. Goiri et al. Sherlock: reliable and efficient agentic workflow execution. arXiv:2511.00330 (2025).
- [48] A. Clark, B. Alomair, L. Bushnell, R. Poovendran. *Submodularity in Dynamics and Control of Networked Systems*. Springer (2016).
- [49] M. Pirani, S. Sundaram. On the smallest eigenvalue of grounded Laplacian matrices. *IEEE Trans. Autom. Control* 61 (2016).
- [50] M. Pirani, E. Moradi Shahrivar, B. Fidan, S. Sundaram. Robustness of leader-follower networked dynamical systems. arXiv:1604.08651 (2016).
- [51] G.L. Nemhauser, L.A. Wolsey, M.L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Math. Program.* 14 (1978).
- [52] J. Thorne, A. Vlachos, C. Christodoulopoulos, A. Mittal. FEVER: a large-scale dataset for fact extraction and verification. NAACL 2018. arXiv:1803.05355.
- [53] P. Lewis et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 2020. arXiv:2005.11401.
- [54] R. Olfati-Saber, R.M. Murray. Consensus problems in networks of agents with switching topology and time-delays. *IEEE Trans. Autom. Control* 49 (2004) 1520–1533.
- [55] X.F. Wang, G. Chen. Pinning control of scale-free dynamical networks. *Physica A* 310 (2002) 521–531.
- [56] H. Chen, W. Ji, L. Xu, S. Zhao. Multi-agent consensus seeking via large language models. arXiv:2310.20151 (2023).
- [57] H. Zhang et al. Stop overvaluing multi-agent debate: we must rethink evaluation and embrace model heterogeneity. arXiv:2502.08788 (2025).

Appendix A Proof of Proposition 2 (oscillatory boundary)

Fix $0 < a < 1$. At $\beta = 0$ the roots of $P(z) = z^\delta(z - a) + \beta$ are 0 and a , strictly inside the unit disk. For $z = e^{i\theta}$, $0 \leq \theta \leq \pi$, put $\phi_a(\theta) = \arg(e^{i\theta} - a)$, taking the continuous argument from 0 to π . Then

$$\phi'_a(\theta) = \frac{1 - a \cos \theta}{1 - 2a \cos \theta + a^2} > 0, \quad \beta = |e^{i\theta} - a| = \sqrt{1 + a^2 - 2a \cos \theta}.$$

Positive β requires $\delta\theta + \phi_a(\theta) = (2j+1)\pi$. The left side is strictly increasing; the smallest admissible angle $\theta_\star$ is its unique solution at $j = 0$. Since $\partial\phi_a/\partial a = \sin \theta / (1 - 2a \cos \theta + a^2) > 0$ for $0 < \theta < \pi$, the limiting arguments at $a = 0$ and $a = 1$ give $\theta < \phi_a(\theta) < (\pi + \theta)/2$. Hence $\pi/(2\delta+1) < \theta_\star < \pi/(\delta+1)$. The modulus above increases strictly with θ , so this angle and its conjugate give the first crossing; any other admissible angle, including $\theta = \pi$ when allowed, has larger dose. Separating real and imaginary parts of $P(e^{i\theta_\star}) = 0$ gives (3).

Every unit-circle root is simple, since $P'(z) = z^{\delta-1}((\delta+1)z - \delta a)$ cannot vanish there. Implicit differentiation at such a root yields

$$\frac{d \log |z|}{d\beta} = \Re \frac{z - a}{\beta((\delta+1)z - \delta a)} = \frac{\delta + 1 + \delta a^2 - (2\delta + 1)a \cos \theta}{\beta |(\delta+1)z - \delta a|^2} > 0.$$

Indeed, the numerator is at least $(1-a)(\delta+1-\delta a) > 0$. All crossings are outward, so stability cannot return at larger β . This proves the stated necessary-and-sufficient criterion for every integer $\delta \geq 1$, without a finite-delay enumeration.

Appendix B Proof of Lemma 2 (monotonicity of the dose)

Write $a_\delta(c) = U_\delta/U_{\delta-1}$. The Chebyshev recurrence $U_\delta = 2cU_{\delta-1} - U_{\delta-2}$ gives a continued fraction and the derivative recursion

$$a_\delta = 2c - \frac{1}{a_{\delta-1}}, \quad a'_\delta = 2 + \frac{a'_{\delta-1}}{a_{\delta-1}^2}.$$

For $\delta \geq 2$, the relevant branch has $c > \cos(\pi/\delta)$, to the right of the largest zero of $U_{\delta-1}$ and of all preceding polynomials. Thus the denominators in this induction are nonzero. Since $a'_1 = 2$, it follows that $a'_\delta > 0$ on this branch. All zeros of $U_{\delta-1}$ are smaller than c , and its leading coefficient is positive; the product representation therefore gives $U_{\delta-1} > 0$ and $U'_{\delta-1} > 0$. Hence

$$\frac{d\beta_c}{da} = -\frac{U'_{\delta-1}}{U_{\delta-1}^2 a'_\delta} < 0,$$

which proves strict decrease in the instantaneous coefficient a for $\delta \geq 2$. At $\delta = 1$, $U_0 = 1$ and $\beta_c = 1$, so this derivative is zero.

For delay monotonicity, fix $a \in (0, 1)$ and use the phase equation from Appendix A. If θ_δ solves $\delta\theta + \phi_a(\theta) = \pi$, then $(\delta+1)\theta_\delta + \phi_a(\theta_\delta) = \pi + \theta_\delta > \pi$. Strict increase of the phase implies $\theta_{\delta+1} < \theta_\delta$. Since $\sqrt{1+a^2-2a\cos\theta}$ increases strictly with θ , it follows that $\beta_c(a, \delta+1) < \beta_c(a, \delta)$ for every integer $\delta \geq 1$.

At $a = 0$, direct factorization gives $\beta_c = 1$ for all delays. As $a \uparrow 1$, the phase equation gives $\theta_* \rightarrow \pi/(2\delta+1)$ and $\beta_c(1, \delta) = 2\sin(\pi/(4\delta+2))$. At $\delta = 2$, eliminating c gives $\beta_c^2 + a\beta_c = 1$, hence the positive solution in the lemma.

Appendix C Two-delay locus and equal-delay ceiling

For Theorem 3, substitute $z = e^{i\theta}$ into $P(z) = 0$ and divide by z^h . The real and imaginary parts are (10). Viewed as equations for (p, q) , their coefficient matrix has determinant $\sin((\delta-d)\theta)$. If it is nonzero, Cramer's rule gives the displayed parametrization. If it vanishes, the rank-one system must instead be solved for consistency; it may contribute a line rather than a parametrized point. Direct substitution at $z = 1$ gives $p + q = 0$, and at $z = -1$ gives

$$p(-1)^d + q(-1)^\delta = 2.$$

No other unit-circle roots are possible. Root counts are locally constant away from this locus, by continuity of roots and the fixed leading coefficient of P . Consequently each connected open cell

can be classified by counting roots at one representative point. At $d = 0$, the regular curve agrees with Proposition 2 on its branch $0 < 1 - p < 1$. The equality $d = \delta$ is a separate integer-delay case, not a continuous limit through the singular parametrization.

For Corollary 3, write $b = p + q > 0$, so the polynomial becomes $z^{\tau+1} - z^\tau + b$. At $b = 0$ the root at 1 is simple and has derivative $dz/db = -1$; all other roots are at zero. Thus sufficiently small positive b is stable. For a non-real unit-circle root with $0 < \theta < \pi$, the phase equation is $\tau\theta + (\pi + \theta)/2 = (2j + 1)\pi$ and $b = 2\sin(\theta/2)$. The first admissible angle is $\theta_* = \pi/(2\tau + 1)$, yielding the claimed threshold. At any unit-circle crossing with $b > 0$, the radial derivative from Appendix A with $a = 1$ has numerator $(2\tau + 1)(1 - \cos \theta) > 0$; the roots are simple and cross outward. Stability therefore does not return at higher gain. Taking the largest μ gives the network condition.

Appendix D Proof of Theorem 2 (greedy placement)

Throughout, write $M(R) = G + \kappa I + w \sum_{i \in R} u_i u_i^\top \succ 0$ and $H(R) = \text{tr } M(R)^{-1}$, with the fixed baseline G of Section 5.

Monotonicity. For $i \notin R$, pinning is a positive-semidefinite update,

$$M(R \cup \{i\}) = M(R) + w u_i u_i^\top \succeq M(R) \implies M(R \cup \{i\})^{-1} \preceq M(R)^{-1},$$

so $H(R \cup \{i\}) \leq H(R)$. Thus H is non-increasing and $\rho(R) = H(\emptyset) - H(R)$ is non-decreasing with $\rho(\emptyset) = 0$.

Marginal gain. By the Sherman–Morrison identity,

$$(M + w u_i u_i^\top)^{-1} = M^{-1} - \frac{w M^{-1} u_i u_i^\top M^{-1}}{1 + w u_i^\top M^{-1} u_i},$$

whose trace is $\text{tr } M^{-1} - w \|M^{-1} u_i\|^2 / (1 + w u_i^\top M^{-1} u_i)$. Subtracting recovers (8),

$$\Delta_i(R) = \frac{w \|M(R)^{-1} u_i\|^2}{1 + w u_i^\top M(R)^{-1} u_i} \geq 0.$$

Submodularity. We must show the marginal is non-increasing in the pinned set,

$$\Delta_i(R) \geq \Delta_i(S) \quad \text{for } R \subseteq S, \ i \notin S.$$

Here the structure of G is essential. Because $M(R) = G + \kappa I + W_R$ has nonpositive off-diagonal entries (from $-A$) and is positive definite, it is a symmetric nonsingular M -matrix, so

$$M(R)^{-1} \geq 0 \quad \text{entrywise.}$$

This inverse sign can be checked directly: choose $s > \lambda_{\max}(M)$ and write $B = I - M/s$. Then B is entrywise non-negative, its eigenvalues lie in $[0, 1)$, and $M^{-1} = s^{-1} \sum_{j=0}^{\infty} B^j$ is entrywise non-negative. The same argument applies after any non-negative diagonal update. For a direct diminishing-returns proof, let $Q(x) = (G + \kappa I + \text{diag } x)^{-1}$ and $f(x) = \text{tr } Q(x)$ for $x \geq 0$. Matrix differentiation gives

$$\partial_i Q = -Q u_i u_i^\top Q, \quad \partial_i f = -(Q^2)_{ii}, \quad \partial_j \partial_i f = 2Q_{ij}(Q^2)_{ij} \geq 0.$$

The last sign follows from entrywise non-negativity and symmetry of Q . Thus $-\partial_i f$ is non-increasing as any other pin weight grows. Integrating it from 0 to w along coordinate i proves $\Delta_i(R) \geq \Delta_i(S)$ for $R \subseteq S$, $i \notin S$. The Loewner order alone would not prove this sign for an arbitrary positive-definite matrix. The numerical sweep in `placement_demo.py --sweep` finds 0/4000 violations for Laplacian-plus-diagonal matrices and 115/4000 for arbitrary positive-definite baselines.

Greedy bound. Write S_j for the greedy set after j steps, $S_0 = \emptyset$ in this argument, and let R^* be an optimal size- k soft-pin set. Monotonicity and submodularity give

$$\rho(R^*) - \rho(S_j) \leq \sum_{i \in R^* \setminus S_j} \Delta_i(S_j) \leq k(\rho(S_{j+1}) - \rho(S_j)).$$

Thus the remaining gap after a step is at most $1 - 1/k$ times the previous gap. Iterating from $\rho(\emptyset) = 0$ gives the finite- k bound in (9); $(1 - 1/k)^k \leq e^{-1}$ gives its last inequality [51]. Substituting $\rho(R) = H(\emptyset) - H(R)$ yields the stated bound on H .

Appendix E Reproducibility

Repository: <https://github.com/YehudaItkin/delayed-verification-llm>. For offline reproduction, use Python 3.11 or later and run from the repository root:

```
python -m pip install -r requirements-reproduce.txt
python reproduce.py
```

This checks `RESEARCH_MANIFEST.json`, restores gzip-compressed logs and recomputes the expanded analyses without querying a model. For restoration alone, run `python restore_data.py`. Paths below refer to restored files. Missing historical raw generations still prevent independent reconstruction of some old scores and parser outputs.

Targeted Lean 4 certificates are under `paper/lean/`: `PlacementCheck.lean` covers the placement counterexample and reduction-bound algebra; `DelayCheck.lean` the delay-allocation and singular-locus examples; `ControllerCheck.lean` archived indexing; `RecheckCheck.lean` gossip-only, averaging and identifiability witnesses; and `RereadCheck.lean` the normal directed matrix and fixed-peer expansion. They do not certify the complete stability or placement theorems or the empirical model.

Expanded-study paths are relative to `experiments/factual_expanded_v4/`: `code_v4_3/PR OTOCOL_RU.md` specifies collection; `qwen38/v4_3_snapshot/results/primary/manifest.json` freezes questions, evidence, conditions, model digest and code hashes. The adjacent `requests.jsonl` and `runs.jsonl` preserve all requests, responses and states; `primary_v4_3_verification.json` records the technical replay. Semantic decisions and their application are in `semantic_review_20260915/ (v1)` and `semantic_review_v2_20260915/ (v2)`. `outcome_analysis_20260915/` contains the post-hoc protocol, code, tests, per-question results, hashes and 20000 bootstrap replicates per dataset and version, with seeds 20260915 (PsiloQA) and 20260916 (TruthfulQA). These records verify calculations conditional on frozen labels, not their independent factual validity.